P 4 Control of Excipients

P 4.1 Specifications and Certificate of analysis (of all excipients)

All the excipients used in manufacturing of Solifenacin Succinate Tablets BP 5 mg are compendial and have been analyzed as per the test methods mentioned in the respective monographs.

The excipients used in Solifenacin Succinate Tablets BP 5 mg are enlisted below:

Sr. No.	Ingredients	Specifications
1	Lactose Monohydrate	BP
2	Maize Starch	BP
3	Hypromellose	BP
4	Magnesium Stearate	BP
5	Ready coat 36FO193220	IH
6	Purified Water	BP

BP: British Pharmacopoeia

IH: In House

Specifications of excipients used in manufacturing of Solifenacin Succinate Tablets BP 5 mg are enclosed herewith

MASTER COPY

LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA		Specification No.	LLS/SPC/RM/015/00
			Supersedes	Nil
Quality Control Department	Raw Material Specification		Effective Date	07/12/2020
	Item Name	Solifenacin Succinate BP		
	Reference	BP	Market	Export

(1) Details of Test and Specification:

Sr. No.	Tests	Specifications
1.	Description	White or light yellow powder.
2.	Solubility	Very soluble or freely soluble in water, soluble in ethanol (96 per cent), practically insoluble in heptane.
3.	Identification	
	A. By IR	The spectrum obtained from sample should be concordant with spectrum obtained with solifenacin succinate CRS.
	B. By Isometric purity	Should Comply for the test of isomeric purity.
4.	Isometric purity	
	Impurity F	NMT 0.15%
	Impurity G	NMT 0.15%
	Impurity H	NMT 0.15%
5.	Impurity E by TLC	Any spot due to impurity E is not more intense than the corresponding spot in the chromatogram obtained with the reference solution (0.1 %).
6.	Related substances	
	Unspecified Impurities	NMT 0.10%
	Total Impurities	NMT 0.3%
7.	Loss on drying	NMT 0.5%
8.	Sulfated ash	NMT 0.1%
9.	Assay	99.0% to 101.0% (Dried substance)

Ref. STP No.: LLS/STP/RM/015
Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Name	M. Arun Gajiyani	Syed Iftekhar	Amol patil
Signature			Apatil.
Date	07/12/2020	07/12/2020	07/12/2020
Department	Quality Control	Quality Control	Quality Assurance

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA		Specification No.	LLS/SPC/RM/017/00
			Supersedes	Nil
Quality Control Department	Raw Material Specification		Effective Date	10/12/2020
	Item Name	Lactose Monohydrate BP		
	Reference	BP	Market	Export

(1) Details of Test and Specification:

Sr. No.	Tests	Specifications
1.	**Description**	White or almost white, crystalline powder.
2.	**Solubility**	Freely soluble in water, practically insoluble in ethanol (96 per cent).
3.	**Identification** First identification: A, D Second identification: B, C, D	
	A. By IR	Compare the spectra with lactose monohydrate CRS.
	B. By TLC	The principal spot in the chromatogram obtained with the test solution is similar in position, colour and size to the principal spot in the chromatogram obtained with reference solution (a).
	C. By Reaction	A red colour develops.
	D. By Water	4.5 % to 5.5 %
4.	**Appearance of solution**	Solution S is clear and not more intensely coloured than reference solution BY7.
5.	**Acidity or alkalinity**	Not more than 0.4 mL of 0.1 M sodium hydroxide is required to change the colour of the indicator to pink or red.
6.	**Specific Optical rotation**	+ 54.4 to + 55.9 (anhydrous substance).
7.	**Absorbance**	At 400 nm: maximum 0.04 for test solution (a) From 210 nm to 220 nm: maximum 0.25 for test solution (b) From 270 nm to 300 nm: maximum 0.07 for test solution (b).
8.	**Water**	4.5 % to 5.5 %
9.	**Sulfated ash**	NMT 0.1%
10.	**Microbial contamination**	TAMC: acceptance criterion 100 CFU/g. Escherichia coli : Absent/g

Ref. STP No.: LLS/STP/RM/017
Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Name	M. Amin Gojiyani	Syed Iffesham	Amol patil
Signature			Apatil
Date	10/12/2020	10/12/2020	10/12/2021
Department	Quality Control	Quality Control	Quality Assurance

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA		Specification No.	LLS/SPC/RM/020/00
			Supersedes	Nil
Quality Control Department	Raw Material Specification		Effective Date	12/12/2020
	Item Name	Maize Starch BP		
	Reference	BP	Market	Export

(1) Details of Test and Specification:

Sr. No.	Tests	Specifications
1.	**Description**	Matt, white to slightly yellowish, very fine powder that creaks when pressed between the fingers.
2.	**Solubility**	Practically insoluble in cold water and in ethanol (96 %)
3.	**Identification**	
	A) By Microscope Examination	Polyhedral granules, 2 to 23 µm in size, or rounded granules, 25 to 32 µm in diameter. The central hilum consists of a distinct cavity or two- to five-rayed cleft no concentric striations. Viewed between crossed nicol prisms, a distinct black cross is seen intersecting at the hilum.
	B) By Physical	Cloudy mucilage is formed.
	C) By Chemical	An orange-red to dark blue colour is produced, which disappears on heating.
4.	**pH**	4.0 to 7.0
5.	**Foreign matter**	Not more than traces of matter other than starch granules are present. No starch grains of any other origin are present.
6.	**Oxidizing Substances**	Not more than 20 ppm. (Calculated as H_2O_2).
7.	**Sulfur Dioxide**	Not more than 50 ppm.
8.	**Iron**	Not more than 10 ppm.
9.	**Loss On Drying**	Not more than 15.0%.
10.	**Sulphated Ash**	Not more than 0.6%.
11.	**Microbial Contamination**	TAMC: 1000 cfu/gm. TYMC: 100 cfu/gm. Absence of Escherichia coli. Absence of Salmonella.

Ref. STP No.: LLS/STP/RM/020

Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Name	Akash C. Nighot	Syed Iffekhar	Amol patil
Signature			Apatil.
Date	15/12/2020	12/12/2020	12/12/2020
Department	Quality Control	Quality Control	Quality Assurance

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola- 444104 (MS) INDIA		Specification No.	LLS/SPC/RM/022/00
			Supersedes	Nil
Quality Control Department	Raw Material Specification		Effective Date	14/12/2020
	Item Name	Hypromellose BP		
	Reference	BP	Market	Export

(1) Details of Test and Specification:

Sr. No.	Tests	Specifications
1.	Description	White, yellowish-white or greyish-white powder or granules, hygroscopic after drying.
2.	Solubility	Practically insoluble in hot water, in acetone, in anhydrous ethanol and in toluene. It dissolves in cold water giving a colloidal solution.
3.	Identification	
	A. By Chemical	The powdered material aggregates on the surface.
	B. By Chemical	A clear or slightly turbid solution occurs with its thickness dependent on the viscosity grade.
	C. By Chemical	A red colour develops at first and changes to purple within 100 min.
	D. By Physical	A coherent, clear film forms on the glass slide.
	E. By Chemical	The flocculation temperature is higher than 50 °C.
4.	Appearance of solution	The solution is not more opalescent than reference suspension III and not more intensely coloured than reference solution Y_6.
5.	pH	5.0 to 8.0.
6.	Viscosity	600 to less than 1400 mPa.s
7.	Loss on Drying	NMT 5.0%
8.	Sulfated Ash	NMT 1.5%
9.	Assay	
	Methoxy Content	28% to 30%
	Hydroxypropoxy Content	7.0% to 12.0%

Ref. STP No.: LLS/STP/RM/022
Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Name	Akash G. Ninghot	Syed Iqteechar	Amol Patil
Signature			
Date	14/12/2020	14/12/2020	14/12/2020
Department	Quality Control	Quality Control	Quality Assurance

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA	Specification No.	LLS/SPC/RM/023/00
		Supersedes	Nil
Quality Control Department	Raw Material Specification	Effective Date	15/12/2020
	Item Name: Magnesium Stearate BP		
	Reference: BP	Market	Export

(1) Details of Test and Specification:

Sr. No.	Tests	Specifications
1.	**Description**	White or almost white, very fine, light powder, greasy to the touch.
2.	**Solubility**	Practically insoluble in water and in anhydrous ethanol.
3.	**Identification:** First identification: C, D Second identification: A, B, D	
	A) Freezing Point	Not less than 53°C
	B) Acid value	195 to 210
	C) By GC	The 2 principal peaks in the chromatogram obtained with the test solution are similar in retention time to the 2 principal peaks in the chromatogram obtained with the reference solution.
	D) By Chemical Reaction	A white crystalline precipitate is formed.
4.	**Acidity or alkalinity**	Not more than 0.05 ml of 0.1 M hydrochloric acid or 0.1 M NaOH is required to change the colour of the solution.
5.	**Chlorides**	Not more than 0.1%.
6.	**Sulphates**	Not more than 1.0%.
7.	**Cadmium**	Not more than 3 ppm.
8.	**Lead**	Not more than 10 ppm.
9.	**Nickel**	Not more than 5 ppm.
10.	**Loss on Drying**	Not more than 6.0%
11.	**Microbial contamination**	TAMC: acceptance criterion 10^3 CFU/g. TYMC: acceptance criterion 10^2 CFU/g. Absence of Escherichia coli. Absence of Salmonella.

	Prepared By	Reviewed By	Approved By
Name	M. Amin Gajiyani	Syed Afteleban	Amol patil
Signature			Apatil
Date	15/12/2020	15/12/2020	15/12/2020
Department	Quality Control	Quality Control	Quality Assurance

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA		Specification No.	LLS/SPC/RM/023/00
			Supersedes	Nil
Quality Control Department	Raw Material Specification		Effective Date	15/12/2020
	Item Name	Magnesium Stearate BP		
	Reference	BP	Reference	BP

Sr. No.	Tests	Specifications
12.	Assay	A. Magnesium: 4.0 % to 5.0 % dried substance. B. Fatty acid composition. Stearic acid : NLT 40.0 % Sum of stearic acid and palmitic acid : NLT 90.0 %

Ref. STP No.: LLS/STP/RM/023
Remark : As per new format SOP No. SOP/QCD/086

Sign and Date	Prepared By	Reviewed By	Approved By
	15/12/2020	15/12/2020	15/12/2020

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA	Specification No.	LLS/SPC/RM/024/00
		Supersedes	Nil
Quality Control Department	Raw Material Specification	Effective Date	16/12/2020
	Item Name	Readycoat Iron Oxide Yellow 36FO193220	
	Reference	IH	Market: Export

(1) Details of Test and Specification:

Sr. No.	Tests	Specifications
1.	**Description**	Yellow coloured powder.
2.	**Identification**	White coloured film formed.
3.	**Particle Size**	NLT 99% pass through 100# ASTM sieve.
4.	**Loss on Drying At (110 °C for 2h)**	NMT 10.0 %.
5.	**Arsenic (ppm)**	NMT 3 ppm.
6.	**Tapped Density**	0.25 g/ml to 0.80 g/ml.

Ref. STP No.: LLS/STP/RM/024
Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Name	Mr. Arjun Gajiyani	Syed Iftekhar	Amol Patil
Signature			@patil
Date	16/12/2020	16/12/2020	16/12/2020
Department	Quality Control	Quality Control	Quality Assurance

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA	Specification No.	LLS/SPC/WS/001/00
		Supersedes	Nil
Quality Control Department	Specification Water Sample	Effective Date	13/01/2021
	Item Name: Purified Water		
	Reference: IH/IP/BP/USP	Market	Export

(1) Details of Test and Specification:

Sr. No.	Tests	Specifications
1	Description	Clear colorless, odorless liquid.
2	pH	Between 5.0 to 7.0
3	Sulphate	Any Opalescence produced in the test solution is not more intense than in the standard solution.
4	Chloride	Any Opalescence produced in the test solution is not more intense than in the standard solution.
5	Ammonium	Yellow colour produced in the test solution should not more intense than in the standard solution.
6	Nitrate	Blue colour produced in the test solution should not more intense than in the standard solution.
7	Heavy Metal	Brown colour observed in test solution is not more intense than that in the Standard solution.
8	Oxidisable substances	The test solution should remain faintly pink.
9	Residue on evaporation	NMT 10 ppm.
10	Conductivity	NMT1.3 µS /cm.
11	TOC	NMT 500 ppb
12	Calcium and Magnesium	A pure blue colour should produce.
13	Carbon Dioxide	The mixture should be clear.
14	Acidity or Alkalinity	Resulting solution should not Red Resulting solution should not Blue
15	Microbial Limit Test	
	Total Viable Aerobic Count	NMT 100 cfu / ml.
	E. coli	Should Absent/100 ml
	Salmonella spp.	Should Absent/100 ml
	P. aeruginosa	Should Absent/100 ml
	S. aureus	Should Absent/100 ml

	Prepared By	Reviewed By	Approved By
Name	Sanjog Deshmukh	Sagar Ithalekar	Amol patil
Signature			@patil
Date	13/01/2021	13/01/2021	18/01/2021
Department	Quality Control	Quality Control	Quality Assurance

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA	Specification No.	LLS/SPC/WS/001/00
		Supersedes	Nil
Quality Control Department	Specification Water Sample	Effective Date	13/01/2021
	Item Name: Purified Water		
	Reference: IH/IP/BP/USP	Market	Export

Note: 1. In case of failure of TOC Analyzer, test for Oxidizable Substances shall be perform.
2.Test of conductivity and TOC shall be performed only for sampling point SP-304 (In-return line of Purified Water loop).

Ref. STP No.: LLS/STP/WS/001
Remark : As per new format SOP No. SOP/QCD/086

Sign and Date	Prepared By	Reviewed By	Approved By
	13/01/2021	13/01/2021	13/01/2021

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Certificate of analysis

The excipients used in Solifenacin Succinate Tablets BP 5 mg are enlisted below:

Sr. No.	Ingredients	Specifications
1	Lactose Monohydrate	BP
2	Maize Starch	BP
3	Hypromellose	BP
4	Magnesium Stearate	BP
5	Ready coat 36FO193220	IH
6	Purified Water	BP

BP: British Pharmacopoeia

IH: In House

Certificate of analysis of excipients used in manufacturing of Solifenacin Succinate Tablets BP 5 mg are enclosed herewith

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Solifenacin Succinate BP			
Batch No.	:	SS1/010/2021	**AR No.**	:	L21RM00085
Mfg. Date	:	02/10/2020	**Exp. Date**	:	01/09/2025
Date of Sampling	:	24/12/2020	**Date of Release**	:	01/01/2021

Sr. No.	Test	Specification	Observation
1.	**Description**	White or light yellow powder.	White powder
2.	**Solubility**	Very soluble or freely soluble in water, soluble in ethanol (96 per cent), practically insoluble in heptane.	Very soluble or freely soluble in water, soluble in ethanol (96 per cent), practically insoluble in heptane.
3.	**Identification**		
	A. By IR	The spectrum obtained from sample should be concordant with spectrum obtained with solifenacin succinate CRS.	Complies
	B. By Isometric purity	Should Comply for the test of isomeric purity.	Complies
4.	**Isometric purity**		
	Impurity F	NMT 0.15%	0.03%
	Impurity G	NMT 0.15%	0.04%
	Impurity H	NMT 0.15%	0.07%
5.	**Impurity E by TLC**	Any spot due to impurity E is not more intense than the corresponding spot in the chromatogram obtained with the reference solution (0.1 %).	Complies
6.	**Related substances**		
	Unspecified Impurities	NMT 0.10%	0.02%
	Total Impurities	NMT 0.3%	0.13%

	Prepared By	Reviewed By	Approved By
Name	Akash C. Ninghot	Syed Iftekhar	Amol patil
Signature			@patil
Date	01/01/2021.	01/01/2021	01/01/2021
Department	Quality Control	Quality Control	Quality Assurance

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Solifenacin Succinate BP			
Batch No.	:	SS1/010/2021	**AR No.**	:	L21RM00085
Mfg. Date	:	02/10/2020	**Exp. Date**	:	01/09/2025
Date of Sampling	:	24/12/2020	**Date of Release**	:	01/01/2021

Sr. No.	Test	Specification	Observation
7.	**Loss on drying**	NMT 0.5%	0.24%
8.	**Sulfated ash**	NMT 0.1%	0.04%
9.	**Assay**	99.0% to 101.0% (Dried substance)	99.7%

Conclusion: The Material **Complies / ~~Does not comply~~** as per above Specification.
Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Sign and Date	01/01/2021.	01/01/2021	01/01/2021

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Lactose Monohydrate BP			
Batch No.	:	PR/P/LAC-0003/21	**AR No.**	:	L21RM00091
Mfg. Date	:	01/09/2020	**Exp. Date**	:	31/08/2025
Date of Sampling	:	24/12/2020	**Date of Release**	:	02/01/2021

Sr. No.	Tests	Specifications	Observations
1.	**Description**	White or almost white, crystalline powder.	White crystalline powder.
2.	**Solubility**	Freely soluble in water, practically insoluble in ethanol (96 per cent).	Freely soluble in water, practically insoluble in ethanol (96 per cent).
3.	**Identification**		
	A. By IR	Compare the spectra with lactose monohydrate CRS.	The IR Spectrum of test sample matches with the anhydrous lactose CRS
	B. By Water	4.5 % to 5.5 %	5.1779%
4.	**Appearance of solution**	Solution S is clear and not more intensely coloured than reference solution BY7.	Solution S is clear and not more intensely coloured than reference solution BY7.
5.	**Acidity or alkalinity**	Not more than 0.4 mL of 0.1 M sodium hydroxide is required to change the colour of the indicator to pink or red.	0.2mL of 0.1 M sodium hydroxide is required to change the colour of the indicator to pink.
6.	**Specific Optical rotation**	+ 54.4 to + 55.9 (anhydrous substance).	+ 54.6
7.	**Absorbance**	At 400 nm: maximum 0.04 for test solution (a) From 210 nm to 220 nm: maximum 0.25 for test solution (b) From 270 nm to 300 nm: maximum 0.07 for test solution (b).	At 400 nm: maximum 0.023 From 210 nm to 220 nm: maximum 0.102 From 270 nm to 300 nm: maximum 0.019
8.	**Water**	4.5 % to 5.5 %	5.1779%
9.	**Sulfated ash**	NMT 0.1%	0.05%
10.	**Microbial contamination**	TAMC: acceptance criterion 100 CFU/g. Escherichia coli : Absent/g	TAMC:10 CFU/g Escherichia coli : Absent/g

Conclusion: The Material **Complies / ~~Does not comply~~** as per above Specification.

Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Name	Akash G. Nighut	Syed Ifteshar	Amol patel
Signature			
Date	02/01/2021.	02/01/2021	02/01/2021
Department	Quality Control	Quality Control	Quality Assurance

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Maize Starch BP			
Batch No.	:	MS147/2021	**AR No.**	:	L21RM00095
Mfg. Date	:	04/10/2020	**Exp. Date**	:	03/09/2025
Date of Sampling	:	23/12/2020	**Date of Release**	:	04/01/2021

Sr. No.	Test	Specification	Observation
1.	**Description**	Matt, white to slightly yellowish, very fine powder that creaks when pressed between the fingers.	White to slightly yellowish, very fine powder that creaks when pressed between the fingers.
2.	**Solubility**	Practically insoluble in cold water and in ethanol (96 %)	Practically insoluble in cold water and in ethanol (96 %)
3.	**Identification**		
	A) By Microscope Examination	Polyhedral granules, 2 to 23 µm in size, or rounded granules, 25 to 32 µm in diameter. The central hilum consists of a distinct cavity or two- to five-rayed cleft no concentric striations. Viewed between crossed nicol prisms, a distinct black cross is seen intersecting at the hilum.	Complies
	B) By Physical	Cloudy mucilage is formed.	Complies
	C) By Chemical	An orange-red to dark blue colour is produced, which disappears on heating.	Complies
4.	**pH**	4.0 to 7.0	6.13
5.	**Foreign matter**	Not more than traces of matter other than starch granules are present. No starch grains of any other origin are present.	Complies
6.	**Oxidizing Substances**	Not more than 20 ppm. (Calculated as H_2O_2).	Complies
7.	**Sulfur Dioxide**	Not more than 50 ppm.	25.6 ppm
8.	**Iron**	Not more than 10 ppm.	Complies
9.	**Loss On Drying**	Not more than 15.0%.	5.7%

	Prepared By	Reviewed By	Approved By
Name	Akash G. Nighot	Syed Iftekhar	Amol patil
Signature			
Date	04/01/2021	04/01/2021	04/01/2021
Department	Quality Control	Quality Control	Quality Assurance

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Maize Starch BP			
Batch No.	:	MS147/2021	**AR No.**	:	L21RM00095
Mfg. Date	:	04/10/2020	**Exp. Date**	:	03/09/2025
Date of Sampling	:	23/12/2020	**Date of Release**	:	04/01/2021

Sr. No.	Test	Specification	Observation
10.	**Sulphated Ash**	Not more than 0.6%.	0.37%
11.	**Microbial Contamination**	TAMC: 1000 cfu/gm. TYMC: 100 cfu/gm. Absence of Escherichia coli. Absence of Salmonella.	TAMC: Less Than 10 cfu/gm TYMC: Less Than 10 cfu/gm. Escherichia coli: Absent/g Salmonella: Absent/g

Conclusion: The Material **Complies** / ~~**Does not comply**~~ as per above Specification.

Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Sign and Date	04/01/2021	04/01/2021	@patel 04/01/2021

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Hypromellose BP			
Batch No.	:	HM20-210301F21	**AR No.**	:	L21RM00098
Mfg. Date	:	01/09/2020	**Exp. Date**	:	31/08/2025
Date of Sampling	:	31/12/2020	**Date of Release**	:	07/01/2021

Sr. No.	Test	Specification	Observation
1.	**Description**	White, yellowish-white or greyish-white powder or granules, hygroscopic after drying.	White, yellowish-white or greyish-white powder or granules, hygroscopic after drying.
2.	**Solubility**	Practically insoluble in hot water, in acetone, in anhydrous ethanol and in toluene. It dissolves in cold water giving a colloidal solution.	Practically insoluble in hot water, in acetone, in anhydrous ethanol and in toluene. It dissolves in cold water giving a colloidal solution.
3.	**Identification**		
	A. By Chemical	The powdered material aggregates on the surface.	Complies
	B. By Chemical	A clear or slightly turbid solution occurs with its thickness dependent on the viscosity grade.	Complies
	C. By Chemical	A red colour develops at first and changes to purple within 100 min.	Complies
	D. By Physical	A coherent, clear film forms on the glass slide.	Complies
	E. By Chemical	The flocculation temperature is higher than 50 °C.	
4.	**Appearance of solution**	The solution is not more opalescent than reference suspension III and not more intensely coloured than reference solution Y_6.	Complies
5.	**pH**	5.0 to 8.0.	6.8
6.	**Viscosity**	600 to less than 1400 mPa.s	800 mPa.s

	Prepared By	Reviewed By	Approved By
Name	Akash G. Ninghot	Syed Afreen	Amol patil
Signature			
Date	07/01/2021	07/01/2021	07/01/2021
Department	Quality Control	Quality Control	Quality Assurance

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Hypromellose BP			
Batch No.	:	HM20-210301F21	**AR No.**	:	L21RM00098
Mfg. Date	:	01/09/2020	**Exp. Date**	:	31/08/2025
Date of Sampling	:	31/12/2020	**Date of Release**	:	07/01/2021

Sr. No.	Test	Specification	Observation
7.	**Loss on Drying**	NMT 5.0%	1.8%
8.	**Sulfated Ash**	NMT 1.5%	0.7%
9.	**Assay**		
	Methoxy Content	28% to 30%	28.8%
	Hydroxypropoxy Content	7.0% to 12.0%	10.6%

Conclusion: The Material **Complies** / ~~**Does not comply**~~ as per above Specification.
Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Sign and Date	07/01/2021	07/01/2021	07/01/2021

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Magnesium Stearate BP			
Batch No.	:	PR/P/MST-0036/21	**AR No.**	:	L21RM00104
Mfg. Date	:	01/10/2020	**Exp. Date**	:	30/09/2025
Date of Sampling	:	01/01/2021	**Date of Release**	:	09/01/2021

Sr. No.	Test	Specification	Observation
1.	**Description**	White or almost white, very fine, light powder, greasy to the touch.	A White, very fine powder greasy to the touch.
2.	**Solubility**	Practically insoluble in water and in anhydrous ethanol.	Practically insoluble in water and in anhydrous ethanol.
Identification			
3.	**A) By GC**	The 2 principal peaks in the chromatogram obtained with the test solution are similar in retention time to the 2 principal peaks in the chromatogram obtained with the reference solution.	Complies.
	B) By Chemical Reaction	A white crystalline precipitate is formed.	A white crystalline precipitate is formed.
4.	**Acidity or alkalinity**	Not more than 0.05 ml of 0.1 M hydrochloric acid or 0.1 M NaOH is required to change the colour of the solution.	0.02 ml of 0.1 M hydrochloric acid is required to change the colour of the solution.
5.	**Chlorides**	Not more than 0.1%.	Less than 0.1%.
6.	**Sulphates**	Not more than 1.0%.	Less than 1.0%.
7.	**Cadmium**	Not more than 3 ppm.	0.2 ppm
8.	**Lead**	Not more than 10 ppm.	0.04 ppm
9.	**Nickel**	Not more than 5 ppm.	0.05 ppm
10.	**Loss on Drying**	Not more than 6.0%	0.46 %
11.	**Microbial contamination**	TAMC: acceptance criterion 10^3 CFU/g. TYMC: acceptance criterion 10^2 CFU/g. Absence of Escherichia coli. Absence of Salmonella.	TAMC:90CFU/g. TYMC:<10 CFU/g. Escherichia coli: Absent /g Salmonella: Absent /g

	Prepared By	Reviewed By	Approved By
Name	Akash G. Ninghot	Syed Ifteekhar	Amol patil
Signature			
Date	09/01/2021	09/01/2021	09/01/2021
Department	Quality Control	Quality Control	Quality Assurance

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Magnesium Stearate BP			
Batch No.	:	PR/P/MST-0036/21	**AR No.**	:	L21RM00104
Mfg. Date	:	01/10/2020	**Exp. Date**	:	30/09/2025
Date of Sampling	:	01/01/2021	**Date of Release**	:	09/01/2021

Sr. No.	Test	Specification	Observation
			S. aureus :Absent/g P. aeruginosa: Absent/g
12.	Assay	A. Magnesium: 4.0 % to 5.0 % dried substance. B. Fatty acid composition. Stearic acid : NLT 40.0 % Sum of stearic acid and palmitic acid : NLT 90.0 %	Magnesium: 4.2% stearic acid: 54.3 % palmitic acid:100.0%

Conclusion: The Material **Complies** / ~~**Does not comply**~~ as per above Specification.

Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Sign and Date	09/01/2021.	09/01/2021	09/01/2021

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF RAW MATERIAL

Material Name	:	Readycoat Iron Oxide Yellow 36FO193220			
Batch No.	:	SST021103	**AR No.**	:	L21RM00107
Mfg. Date	:	02/10/2021	**Exp. Date**	:	01/09/2026
Date of Sampling	:	04/01/2021	**Date of Release**	:	11/01/2021

Sr. No.	Tests	Specifications	Observations
1.	**Description**	Yellow coloured powder.	Yellow coloured powder
2.	**Identification**	White coloured film formed.	White coloured film formed
3.	**Particle Size**	NLT 99% pass through 100# ASTM sieve.	99.99 % pass through 100# ASTM sieve.
4.	**Loss on Drying At (110 °C for 2h)**	NMT 10.0 %.	1.08%
5.	**Arsenic (ppm)**	NMT 3 ppm.	After 10 min. any stain produced on the mercuric chloride paper is not more intense than that obtained by treating in the same manner.(i.e less than 3 ppm)
6.	**Tapped Density**	0.25 g/ml to 0.80 g/ml.	0.798 g/ml

Conclusion: The Material **Complies / ~~Does not comply~~** as per above Specification.

Remark : As per new format SOP No. SOP/QCD/086

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Date	11/01/2021.	11/01/2021	11/01/2021
Department	**Quality Control**	**Quality Control**	**Quality Assurance**

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF PURIFIED WATER

Material Name	:	PURIFIED WATER			
Batch No.	:	NA	**AR No.**	:	L21PW00120
Mfg. Date	:	NA	**Exp. Date**	:	NA
Date of Sampling	:	01/03/2021	**Date of Release**	:	10/03/2021

Sr. No.	Test	Specification	Observation
1.	Description	Clear colourless, odourless liquid.	Clear colourless, odourless liquid
2.	pH	Between 5.0 to 7.0	6.24
3.	Sulphate	No any opalescence should be produced in test solution.	No any opalescence produced in test solution.
4.	Chloride	No any opalescence should be produced in test solution.	No any opalescence produced in test solution.
5.	Ammonium	The test solution should not more intense than standard solution.	Test solution is not more intense than standard solution.
6.	Nitrate	Blue color in the test solution should not more intense than standard solution.	Blue color in the test solution is not more intense than standard solution.
7.	Heavy Metal	No brown color should be observed in the test solution.	No brown color observed in the test solution.
8.	Oxidisable Substances	The test solution should remain faintly pink.	The test solution remain faintly pink.
9.	Residue on evaporation	NMT 10 ppm.	8.2 ppm
10.	Conductivity	NMT1.3 µS/cm.	0.612 µS/cm.
11.	Calcium and Magnesium	A pure blue colour should produced.	A pure blue colour is produced.
12.	Carbon Dioxide	The solution mixture should be clear.	The test solution mixture is clear.
13.	Acidity and Alkalinity	Resulting solution should not Red Resulting solution should not Blue	Resulting solution not Red Resulting solution not Blue
14.	**Microbial Limit Test**		
	Total Viable Aerobic Count	NMT 100 cfu/ml.	25 cfu/ml.

	Prepared By	Reviewed By	Approved By
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Signature			
Date	10/03/2021	10/03/2021	10/03/2021
Department	Quality Control	Quality Control	Quality Assurance

Leben Life Sciences Pvt. Ltd.

CERTIFICATE OF ANALYSIS OF PURIFIED WATER

Material Name	:	PURIFIED WATER			
Batch No.	:	NA	**AR No.**	:	L21PW00120
Mfg. Date	:	NA	**Exp. Date**	:	NA
Date of Sampling	:	01/03/2021	**Date of Release**	:	10/03/2021

	E. coli	Should Absent/100 ml	Absent
	Salmonella spp.	Should Absent/100 ml	Absent
	P. aeruginosa	Should Absent/100 ml	Absent
	S. aureus	Should Absent/100 ml	Absent

Conclusion: The Material **Complies ~~/ Does not comply~~** as per above Specification.

Remark : As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Sign and Date	10/03/2021	10/03/2021	10/03/2021